

 Marwadi University Marwadi Chandarana Group 	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim: Generate Histogram of images, apply Histogram equalization and Histogram matching on it.	
Experiment No: 8	Date:	Enrollment No: 92510133028

```

import cv2
import matplotlib.pyplot as plt
import numpy as np
from google.colab import files

# Step 1: Upload the image
print("Please upload your image file (e.g., IMG_20230915_135931789.jpg)")
uploaded = files.upload()

if not uploaded:
    print("No file uploaded.")
else:
    # Step 2: Get the filename
    image_path = list(uploaded.keys())[0]
    image = cv2.imread(image_path, cv2.IMREAD_GRAYSCALE)

    if image is None:
        print(f"Error: Could not load image from {image_path}")
    else:
        print(f"Image '{image_path}' loaded successfully.")

    # Calculate and Plot Histogram
    histogram = cv2.calcHist([image], [0], None, [256], [0, 256])
    plt.figure(figsize=(8, 4))
    plt.title('Original Histogram')
    plt.plot(histogram)
    plt.xlim([0, 256])
    plt.show()

    # Perform histogram equalization
    equalized_image = cv2.equalizeHist(image)

    # Display Results
    plt.figure(figsize=(12, 6))
    plt.subplot(1, 2, 1)
    plt.title('Original Image')
    plt.imshow(image, cmap='gray')
    plt.axis('off')

    plt.subplot(1, 2, 2)

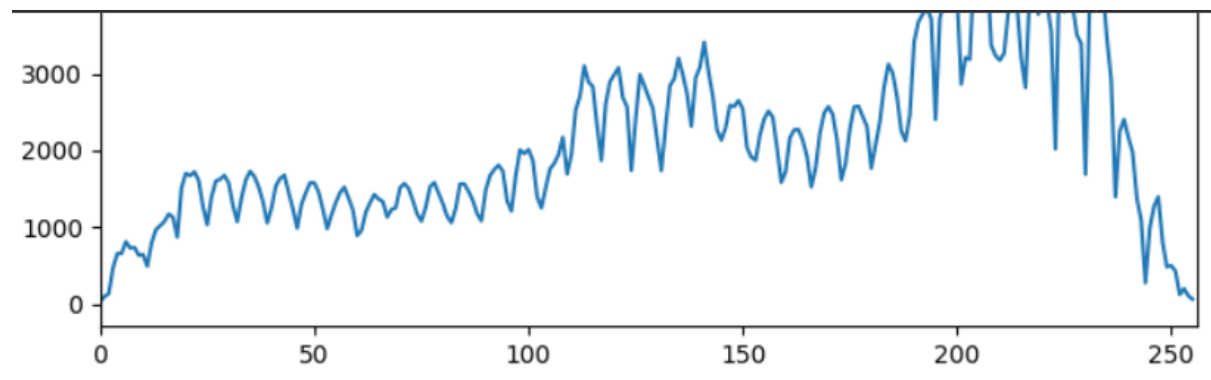
```

 Marwadi University Marwadi Chandarana Group 	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim: Generate Histogram of images, apply Histogram equalization and Histogram matching on it.	
Experiment No: 8	Date:	Enrollment No: 92510133028

```
plt.title('Equalized Image')
plt.imshow(equalized_image, cmap='gray')
plt.axis('off')
plt.show()

# Histogram of Equalized Image
hist_eq = cv2.calcHist([equalized_image], [0], None, [256], [0,
256])

plt.figure(figsize=(8, 4))
plt.title('Equalized Histogram')
plt.plot(hist_eq)
plt.xlim([0, 256])
plt.show()
```



 Marwadi University Marwadi Chandarana Group 	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim: Generate Histogram of images, apply Histogram equalization and Histogram matching on it.	
Experiment No: 8	Date:	Enrollment No: 92510133028

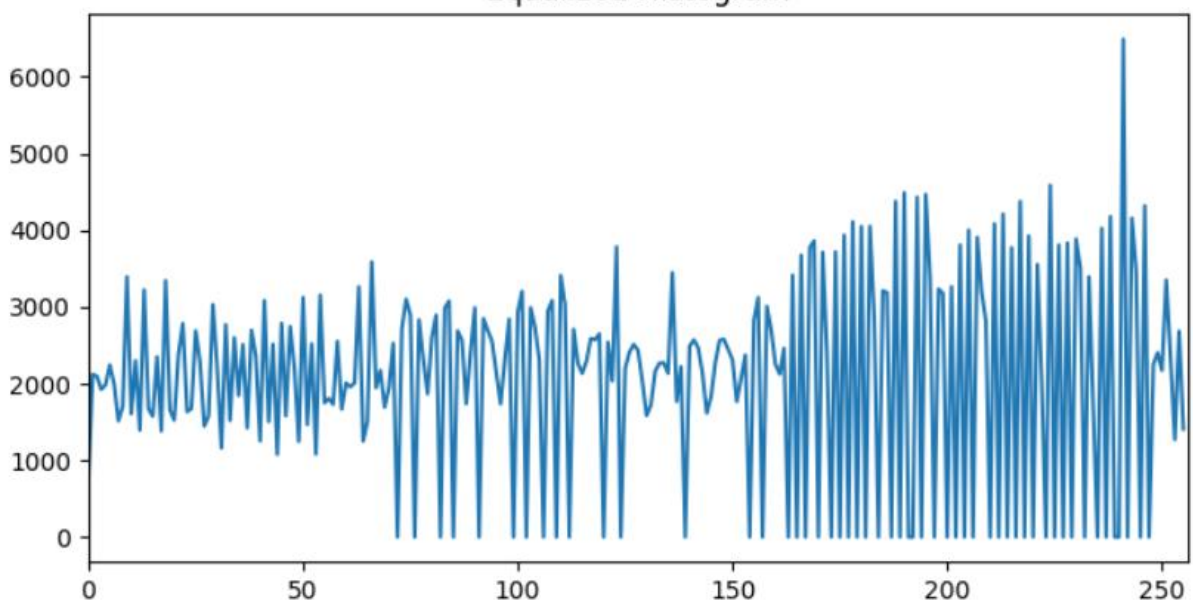
Original Image



Equalized Image



Equalized Histogram



```
import cv2
```

 Marwadi University Marwadi Chandarana Group 	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim: Generate Histogram of images, apply Histogram equalization and Histogram matching on it.	
Experiment No: 8	Date:	Enrollment No: 92510133028

```

import numpy as np
from matplotlib import pyplot as plt
from google.colab import files

# Step 1: Upload the images
print("Please upload the SOURCE image:")
uploaded_source = files.upload()
if not uploaded_source:
    raise ValueError("Source image upload cancelled.")
source_path = list(uploaded_source.keys())[0]

print("\nPlease upload the REFERENCE image:")
uploaded_ref = files.upload()
if not uploaded_ref:
    raise ValueError("Reference image upload cancelled.")
reference_path = list(uploaded_ref.keys())[0]

# Load the source and reference images
source_image = cv2.imread(source_path, cv2.IMREAD_GRAYSCALE)
reference_image = cv2.imread(reference_path, cv2.IMREAD_GRAYSCALE)

if source_image is None or reference_image is None:
    print("Error: One or both images could not be loaded. Please ensure they are valid image files.")
else:
    # Calculate histograms
    source_hist = cv2.calcHist([source_image], [0], None, [256], [0, 256])
    reference_hist = cv2.calcHist([reference_image], [0], None, [256], [0, 256])

    s_sum = source_hist.sum()
    r_sum = reference_hist.sum()

    if s_sum == 0 or r_sum == 0:
        print("Error: One of the images is empty or has no intensity variation.")
    else:
        # Normalize histograms
        source_hist /= s_sum
        reference_hist /= r_sum

```

 Marwadi University Marwadi Chandarana Group 	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim: Generate Histogram of images, apply Histogram equalization and Histogram matching on it.	
Experiment No: 8	Date:	Enrollment No: 92510133028

```
# Calculate cumulative distribution functions (CDF)
source_cdf = source_hist.cumsum()
reference_cdf = reference_hist.cumsum()

# Perform histogram matching
mapping = np.interp(source_cdf, reference_cdf, range(256))
matched_image = mapping[source_image].astype(np.uint8)

# Display the images
plt.figure(figsize=(15, 5))

plt.subplot(131)
plt.title('Source Image')
plt.imshow(source_image, cmap='gray')
plt.axis('off')

plt.subplot(132)
plt.title('Reference Image')
plt.imshow(reference_image, cmap='gray')
plt.axis('off')

plt.subplot(133)
plt.title('Matched Image')
plt.imshow(matched_image, cmap='gray')
plt.axis('off')

plt.tight_layout()
plt.show()
```



 Marwadi University Marwadi Chandarana Group 	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim: Generate Histogram of images, apply Histogram equalization and Histogram matching on it.	
Experiment No: 8	Date:	Enrollment No: 92510133028

Conclusion

From this experiment, I learned how to generate the histogram of an image and understand its intensity distribution. I also learned how histogram equalization improves image contrast and how histogram matching changes an image to have a similar intensity distribution as a reference image. Overall, this experiment helped me understand the practical use of histogram-based image enhancement techniques.

https://github.com/keshvi1234/DSIP_Exp